

Problem-Solution Fit

Date	2 July 2026
Team ID	
Project Name	Voice based concept understanding analyser
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template from 1052.jpg.

1. CUSTOMER SEGMENT(S) [CS] • Independent Students: Learners trying to master complex college coursework online. • Technical Job Aspirants: Candidates preparing for professional viva-voce rounds or interviews.	6. CUSTOMER LIMITATIONS [CL] <i>E.G., BUDGET, DEVICES</i> • Limited student budget restricts premium software purchases. • Varying ambient classroom background noises. • Varying microphonic qualities across low-cost hardware devices.	5. AVAILABLE SOLUTIONS [AS] <i>PROS & CONS</i> • Static MCQ Tests: Easy to automate but completely fail to evaluate expressive spoken vocabulary. • Manual Mock Interviews: Highly targeted feedback but lacks scalability and expensive to afford.
2. PROBLEMS / PAINS [PR] <i>+ ITS FREQUENCY</i> • Blind Spots: Difficulty identifying specific gaps in technical descriptions (Frequency: Constant/Daily). • No Vocal Grading: Inability to test spoken articulation speed and keyword mastery automatically (Frequency: Weekly).	9. PROBLEM ROOT / CAUSE [RC] • Traditional digital tools rely on text inputs, ignoring oral concept validation. • LMS tools focus on binary grading rules instead of vector semantic keyword contexts.	7. BEHAVIOR [BE] <i>+ ITS INTENSITY</i> • Students re-read textbook glossaries loudly to practice lines (Intensity: High). • Candidates perform self-recording on phones to self-evaluate performance (Intensity: Moderate).

Problem-Solution Fit Canvas (Continued)

Strategic mapping layout showcasing behavioral channels, emotional indicators, and the integrated conceptual solution response framework.

3. TRIGGERS TO ACT [TR] • Getting bad results in dynamic mock viva interviews. • Struggling to memorize core keyword strings for exams. • Approaching engineering interview cycles.	10. YOUR SOLUTION [SL] Voice-based Concept Understanding Analyser: An interactive software engine utilizing the OpenAI Whisper pipeline to record student vocals, map semantic vector definitions instantly, highlight missing textbook nodes via visual charts, and offer direct remedial guidance templates.	8. CHANNELS OF BEHAVIOR [CH] ONLINE: • Web-based study forums and LMS boards. • Digital mobile voice notes. OFFLINE: • In-person study groups and traditional classroom lecture tests.
4. EMOTIONS [EM] <i>BEFORE / AFTER</i> • Before: Anxious, confused, and insecure regarding actual verbal fluency parameters. • After: Relieved, self-assured, motivated, and structurally guided.		